ER: Requirements Specification Component

> The book industry has been rapidly shifting towards digital platforms, with readers increasingly purchasing both physical and digital books online. However, many existing platforms lack personalisation, simplicity, and a community-oriented approach. Liberato emerges as an online bookshop designed to combine accessibility, modern technology, and user engagement, making it easier for readers to find, purchase, and enjoy books from anywhere.

A1: Liberato

> Goals and description

Provide an intuitive and secure platform for purchasing books. Support authors and publishers by offering visibility and integration with the platform. Liberato is an online bookstore that offers a wide catalogue of books. The system will provide features such as advanced search, personalised recommendations, user reviews, and a simple checkout process. Beyond being a sales platform, Liberato aims to build a space where readers, students, and authors can connect through literature.

> Main Features

- Online catalogue with advanced search and filtering.
- User accounts with wishlists and favourites.
- Reviews and ratings for books.
- Promotional campaigns and discounts.

> User profiles

- Reader: searches for books to buy and expects a simple, fast experience.
- Seller: can shop as a user but has the permission to put their own books for sale.
- Admin: manages catalogue, users, orders, and promotional campaigns.

A2: Actors and User stories

> The goal of this artifact is to identify and describe the different types of users (actors) who will interact with the system, along with the user stories that capture their needs, motivations, and interactions.
> By defining actors and their corresponding user stories, this artifact provides a clear understanding of system functionalities from a user-centered perspective. It helps ensure that the system design aligns with real-world usage scenarios, guiding both functional requirements and development priorities.

1. Actors

Actor	Relationship	Description
User	Parent / General actor	Represents any person interacting with the system.
Unauthenticated User	Inherits from User	Represents a visitor who can browse products, view details, and read general information but cannot make purchases.
Authenticated User	Inherits from User	Represents a registered user who can log in, manage their profile, make purchases, and access personalised features.
Administrator	Independent of User hierarchy	Represents a separate actor responsible for managing users, products and categories, and maintains the platform.

Actor	Relationship	Description
Seller	Inherits from Authenticated User	Represents a type of authenticated user with additional privileges to add, update, and manage their products and stock.

2. User Stories

> User stories organised by actor.
> For each actor, a table containing a line for each user story, and for each user story: an identifier, a name, a priority, and a description.

2.1 User (unauthenticated)

Identifier	Name	Priority	Responsible	Description
US01	Registration	High	Ricardo	As a visitor, I want to register a new account, so that I can start using the system.
US02	Login	High	João	As a user, I want to log in to my account, so that I can securely access and end my sessions.

2.2 User

Identifier	Name	Priority	Responsible	Description
US03	View & Search Products	High	Iara	As a user, I want to view and search products, so that I can browse the store.
US04	Exact Match Search	High	Ricardo	As a user, I want to search with exact matches, so that I can quickly find specific results.
US05	Full-text Search	High	Dinis	As a user, I want to use full-text search, so that I can find results by keywords.
US06	Add item to Cart	High	Ricardo Fernandes	As a user, I want to add items to my cart, so that I can buy my purchases.
US07	Manage Cart	High	Ricardo Fernandes	As a user, I want to update or remove items in my cart, so that I can manage my purchases.
US08	View Product Details	High	Ricardo	As a user, I want to view detailed information about a product, so that I can make informed purchase decisions.
US09	View Product Reviews	Medium	Iara	As a user, I want to view product reviews, so that I can evaluate others' experiences before buying.
US10	Search over Multiple Attributes	Medium	Iara	As a user, I want to search by multiple attributes, so that I can refine results.
US11	Search Filters	Medium	João	As a user, I want to apply filters to my search, so that I can narrow down results easily.
US12	Ordering of Results	Medium	Ricardo	As a user, I want to order search results, so that I can view items by relevance, date, or other criteria.
US13	Browse Product Categories	Medium	João	As a user, I want to browse product categories, so that I can easily find types of products I'm interested in.

Identifier	Name	Priority	Responsible	Description
US14	Faceted Search	Innovation	Dinis	As a user, I want to use faceted search, so that I can browse results with categorised attributes.
US15	Placeholders in Form Inputs	Medium	Iara	As a user, I want to see placeholders in form inputs, so that I understand what information to enter.
US16	About Us	Medium	Dinis	As a visitor, I want to see an About Us section, so that I can learn about the company.
US17	Main Features	Medium	Iara	As a visitor, I want to see the system's main features, so that I can understand what it offers.
US18	Contacts	Medium	João	As a visitor, I want to view contact information, so that I can reach the company if needed.

2.3 Authenticated user

Identifier	Name	Priority	Responsible	Description
US19	View Profile	High	João	As an authenticated user, I want to view my profile information, so that I can check my personal data.
US20	Edit Profile	High	Ricardo	As an authenticated user, I want to edit my profile, so that I can update my information when needed.
US21	Checkout	High	João Santos	As an authenticated user, I want to proceed to checkout, so that I can complete my purchase securely and efficiently.
US22	Review Purchased Product	High	Dinis	As an authenticated user, I want to review a product I purchased, so that I can share my experience with other users.
US23	Payment Approved	High	Iara	As an authenticated user, I want to receive a notification when my payment is approved, so that I know my order is being processed.
US24	Recover Password	Medium	Dinis	As an authenticated user, I want to recover my password, so that I can regain access if I forget it.
US25	Delete Account	Medium	Iara	As an authenticated user, I want to delete my account, so that I can permanently remove my data from the system.
US26	Upload/Update Profile Picture	Medium	Dinis	As an authenticated user, I want to upload or update my profile picture, so that my account is personalised.
US27	View Personal Notifications	Medium	Iara	As an authenticated user, I want to view my personal notifications, so that I stay informed about updates.
US28	Contextual Error Messages	Medium	João	As an authenticated user, I want to receive contextual error messages, so that I know how to fix issues while using the system.

Identifier	Name	Priority	Responsible	Description
US29	Contextual Help	Medium	Ricardo	As an authenticated user, I want to access contextual help, so that I can get guidance while using the system.
US30	View Notification List	Medium	Ricardo	As a user, I want to view my notification list, so that I can keep track of system updates.
US31	Mark Notifications as Read	Medium	Dinis	As a user, I want to mark notifications as read, so that I can manage which alerts I've already seen.
US32	Log Out	Medium	Dinis Pinto	As an authenticated user, I want to log out of my account, so that I can ensure my personal data and session remain secure when I leave the system.
US33	View Product Details & Reviews	Medium	João Santos	As an authenticated user, I want to view product details and reviews, so that I can decide what to buy.
US34	Manage Purchases	Medium	Dinis Pinto	As an authenticated user, I want to view my purchase history, so that I can track past orders.
US35	Manage Wishlist	Medium	Iara Brás	As an authenticated user, I want to manage a wishlist, so that I can save products for later.
US36	Write Reviews	Medium	João Santos	As an authenticated user, I want to write and report reviews, so that I can share feedback and keep reviews fair.
US37	Edit Review	Medium	Ricardo Fernandes	As an authenticated user, I want to edit my review, so that I can update my feedback based on new experiences or opinions.
US38	Remove Review	Medium	Iara Brás	As an authenticated user, I want to remove my review, so that I can delete feedback I no longer wish to share.
US39	Track Order	Medium	Dinis	As an authenticated user, I want to track my order status, so that I can know when my purchase will arrive.
US40	Cancel Order	Medium	João Santos	As an authenticated user, I want to cancel my order, so that I can stop the purchase if I change my mind or find an issue.
US41	Change in Order Processing Stage	Medium	Dinis	As an authenticated user, I want to be notified when the processing stage of my order changes, so that I can track its progress.
US42	Product in Wishlist Available	Medium	Ricardo	As an authenticated user, I want to receive a notification when a product in my wishlist becomes available, so that I can purchase it quickly.

Identifier	Name	Priority	Responsible	Description
US43	Product on Cart Price Change	Medium	João	As an authenticated user, I want to be notified when a product in my cart changes price, so that I can decide whether to complete the purchase.
US44	Appeal for Unblock	Innovation	João	As a blocked user, I want to send an unblock appeal, so that I can request access to my account again.

2.4 Seller

Identifier	Name	Priority	Responsible	Description
US45	Manage their stock	High	João	As a seller, I want to add, update and manage my products and stock, so that the stock remains accurate and organised.
US46	Manage their sales stats	Medium	Dinis	As a seller, I want to view my sales statistics and manage discounts on my products, so that I can optimise my exposure.

2.4 Admin

Identifier	Name	Priority	Responsible	Description
US47	Support Admin Accounts	High	Iara	As an admin, I want to have a dedicated admin account, so that I can manage system functionalities securely.
US48	Administer User Accounts	High	João	As an admin, I want to search, view, edit, and create user accounts, so that I can manage the platform's users.
US49	Add Product	High	João	As an admin, I want to add new products to the system, so that I can expand the catalog available to users.
US50	Manage Products Information	High	Ricardo	As an admin, I want to edit and update product information, so that all product details remain accurate and up to date.
US51	Manage Order Status	High	João	As an admin, I want to update and manage the status of orders, so that I can oversee order processing and fulfilment.
US52	Block/Unblock User Accounts	Medium	Ricardo	As an admin, I want to block or unblock accounts, so that I can enforce rules and restore access if needed.
US53	Delete User Account (Admin)	Medium	Dinis	As an admin, I want to delete user accounts, so that I can remove inactive or problematic accounts.
US54	Manage Products Stock	Medium	Dinis	As an admin, I want to manage product stock levels, so that I can ensure product availability and prevent overselling.
US55	Manage Product Categories	Medium	Iara	As an admin, I want to create, edit, and delete product categories, so that I can organise the catalog effectively.
US56	View Users' Purchase History	Medium	Ricardo	As an admin, I want to view users' purchase histories, so that I can monitor activity and assist with support or audits.

3. Supplementary Requirements

> Section including business rules, technical requirements, and restrictions.
> For each subsection, a table containing identifiers, names, and descriptions for each requirement.

3.1. Business rules

Identifier	Name	Priority	Category	Description
BR.01	Administrator Role Separation	High	Access/Security	Administrator accounts are independent of user accounts and cannot buy products (i.e., they have no purchasing capability).
BR.02	Product Price Floor	High	Finance/Product	The price of any book on the platform must be greater than 0 (>0).
BR.03	Inventory Pre-Requisite	High	Sales/Inventory	Buyers can only complete a purchase for books that are currently in stock (Stock ≥ 1).
BR.04	Product Data Completeness	High	Data Integrity	Sellers must input the book name and its author as mandatory fields when listing a book for sale.
BR.05	Price Lock During Payment	High	Finance/Transaction	Books that are part of an order currently in the payment processing stage cannot have their price changed.
BR.06	Product Purchase Availability	High	Products/Store Management	A product can only be purchased after its publication date.
BR.07	Order Quantity Limit	Medium	Sales/Inventory	Buyers cannot purchase over X units of the same book in a single order (where X is the available stock for the book).
BR.08	Data Anonymisation	Medium	Privacy/Data Retention	Upon account deletion (FR.014), shared user data (e.g., comments, reviews, likes) must be retained but made permanently anonymous.

3.2. Technical requirements

Identifier	Name	Priority	Category	Description
TR.01	Performance (Response Time)	High	Usability/Speed	The system must have response times shorter than 2 seconds for all critical user actions to ensure sustained user attention and engagement.
TR.02	Robustness (Error Handling)	High	Reliability	The system must be prepared to handle runtime errors gracefully (e.g., database connection failure) and continue operating or fail safely without crashing.
TR.03	Scalability (Growth Management)	High	Maintainability/Growth	The system must be designed to effectively deal with a substantial growth in the number of users and their actions (e.g., horizontal scaling capability).

Identifier	Name	Priority	Category	Description
TR.04	Accessibility (WCAG Compliance)	High	Usability/Legal	The system must ensure that everyone can access and interact with the pages, regardless of disabilities or the specific Web browser they use.
TR.05	Security (Password Storage)	High	Security/Data Integrity	The system must store all user passwords securely, using modern, industry-standard hashing and salting algorithms (e.g., Argon2 or bcrypt).

3.3. Restrictions

Identifier	Name	Category	Description	Justification (Implied)
C.01	Project Deadline	Time/Schedule	The project must be concluded before January 12th (delivery of final working system).	Defines the hard limit for all development and testing phases.
C.02	Version Control System	Tooling/DevOps	GitLab must be used for collaborative software development, documentation, and versioning.	Enforces the entire CI/CD pipeline and code repository location.
C.03	Database System	Technology Stack	PostgreSQL must be used as the primary relational database system.	Restricts the choice of RDBMS, excluding MySQL, MariaDB, etc.
C.04	Backend Language	Technology Stack	PHP must be used as the programming language on the server side.	Enforces the backend language choice.
C.05	Backend Framework	Technology Stack	Laravel must be used as the server web framework.	Enforces the use of a specific PHP framework, limiting architectural freedom.
C.06	Virtualisation Environment	Tooling/Deployment	Docker must be used as the virtualisation environment for deployment.	Enforces containerization for deployment and environment consistency.
C.07	Web Server	Infrastructure	NGINX must be used as the web server (to handle HTTP requests).	Restricts the use of other web servers like Apache HTTP Server.
C.08	Frontend Languages	Technology Stack	HTML, CSS, and JavaScript must be used as client-side languages.	Defines the foundational languages for the browser, common for all web projects.

A3: Information Architecture

> The goal of this artifact is to define the overall structure and organisation of the system's content and navigation. It provides a clear overview of how users will access and interact with the main sections of the platform.

1. Sitemap

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2. Wireframes

> Wireframes for some of the principal pages of our website, the homepage, the product page, the shopping cart and the profile page.

UI01: HomePage

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1. Logo of Liberato.
2. Filter menu, filter items by genre, author, price, language, collection or seller type;
3. Search bar to look through the stock;
4. Browse the most recent available products here;
5. Button to access your notifications, check your latest updates and alerts;
6. Button to access your wishlist, view and manage your saved favorites;
7. Button to access your shopping cart, see the items you plan to buy;
8. Button to access your profile, access and edit your personal account.

UI02: Cart

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1. Book information, displays the book cover, title, and author to identify each item in the cart;
2. Quantity selector, allnnovations you to increase or decrease the number of copies you wish to purchase.

3. Unit price, shows the price for a single copy of the selected book.
4. Remove button, deletes the item from your cart.
5. Cart total, displays the total price of all items in your shopping cart.
6. Checkout button, proceeds to the checkout process to complete the purchase.

UI03: Book Page

1. Product image, displays the book's cover;
2. Main product information, includes book title, author, average rating, and price;
3. Action buttons, add the book to the cart or to your wishlist;
4. Book details section, provides access to synopsis, product specifications (e.g. dimensions, ISBN, language, genre), and customer reviews.

UI04: Profile

1. Displays basic details of the authenticated user, such as name, email, and profile picture.
2. Settings button, allnnovations access to account settings, where the user can view personal data, update passwords, and edit other information.
3. Previous orders summary, provides an overview of previous orders, including order number, purchase date, product details, and total price.

Revision history

Changes made to the first submission:

1. Error correction in user stories;
2. User Stories and requirements organised by priority;
3. Actor system removed, unnecessary;
4. Correction of A1 user profiles

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Comments